



Saint Columban College
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Hidden in plainsight

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Easy Forensics picoMini by CMU-Africa browser_webshell_solvable

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Description

You're given a seemingly ordinary JPG image. Something is tucked away out of sight inside the file. Your task is to discover the hidden payload and extract the flag.

Download the jpg image [here](#).

Hints ?

1

Download the jpg image and read its metadata

24,644 users solved



95% Liked



picoCTF{FLAG}

Submit Flag

Author: The Analyst: Hyposelenia

Challenge: Hidden in plainsight

Category: Forensics

Date: 03/07/26



I. Objective

The objective of this challenge is to analyze an image file and investigate hidden information within it using forensic tools in order to retrieve the concealed flag.

II. Background

In digital forensics and steganography, data can be hidden inside files such as images, audio, or videos without visibly altering the file. This technique is known as **steganography**, where information is concealed within another file to avoid detection.

Images may also contain **metadata**, which stores additional information about the file such as camera details, author information, comments, and other hidden fields. Sometimes these metadata fields contain clues that lead investigators to hidden data.

Tools like **exiftool** allow analysts to inspect metadata, while tools such as **steghide** can extract hidden files embedded inside images.

III. Tool Used

- **Kali Linux (VirtualBox)** – Used as the operating system environment for performing forensic analysis.
- **ExifTool** – Used to read and analyze metadata stored inside the image file.
- **CyberChef** – Used to decode Base64 encoded strings.
- **Steghide** – Used to extract hidden data embedded within the image file.

IV. Methodology

1. Launched **Kali Linux** using VirtualBox.



2. Downloaded the image file provided in the challenge description.

the file. Your task is to discover th

Download the jpg image [here](#).

3. Used the command below to inspect the metadata of the image:

```
File Actions Edit View Help
(kali@kali)-[~/Downloads]
$ ls
img.jpg
(kali@kali)-[~/Downloads]
$ exiftool img.jpg
```

exiftool img.jpg

4. Observed the metadata output and noticed an **unusual encoded string in the Comment field**.

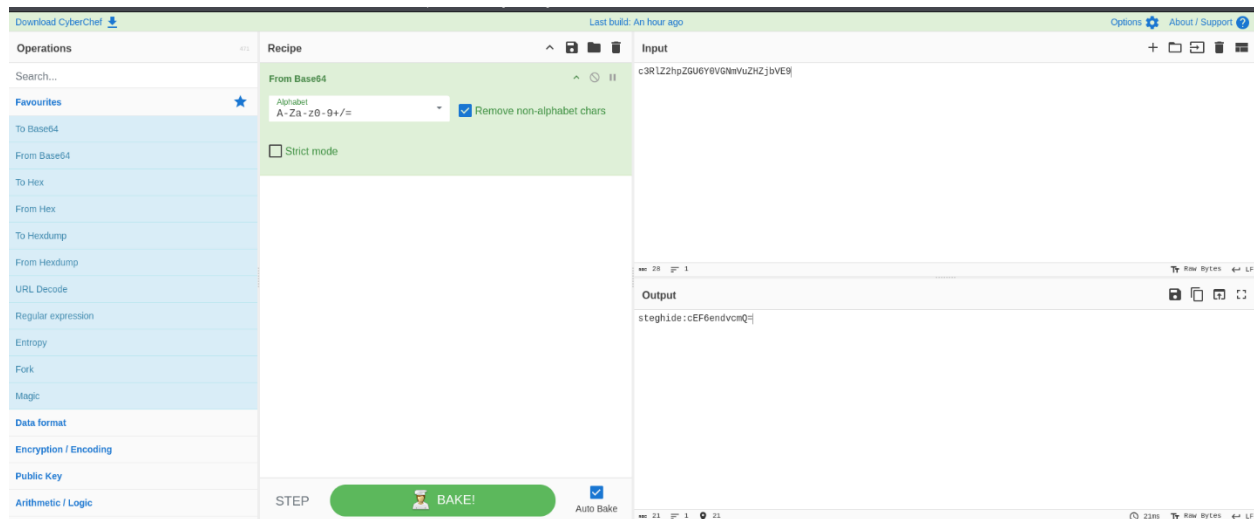
```
(kali@kali)-[~/Downloads]
$ exiftool img.jpg
ExifTool Version Number      : 13.25
File Name                    : img.jpg
Directory                    : 
File Size                    : 74 kB
File Modification Date/Time   : 2026:03:07 08:26:04-05:00
File Access Date/Time        : 2026:03:07 08:26:04-05:00
File Inode Change Date/Time   : 2026:03:07 08:26:04-05:00
File Permissions              : -rw-rw-r--
File Type                    : JPEG
File Type Extension          : jpg
MIME Type                    : image/jpeg
JFIF Version                 : 1.01
Resolution Unit              : None
X Resolution                  : 1
Y Resolution                  : 1
Comment                      : c3RlZ2hpZGU6Y0VGbWVuZHZjbVE9
Image Width                  : 640
Image Height                  : 640
Encoding Process              : Baseline DCT, Huffman coding
Bits Per Sample              : 8
Color Components              : 3
Y Cb Cr Sub Sampling         : YCbCr4:2:0 (2 2)
Image Size                   : 640x640
Megapixels                   : 0.410
```



5. Copied the encoded text and opened **CyberChef** to analyze it.

```
Y Resolution : 1  
Comment : c3RlZ2hpZGU6Y0VGNmVuZHZjbVE9  
Image Width : 640  
Image Height : 640
```

6. Used the **Base64** decoding operation.



7. After decoding once, the output revealed the word “**steghide**”.

```
Output  
steghide:cEF6endvcmQ=
```

8. Decoded again and discovered the word “**pAzzword**”, which appeared to be a password.

```
Output  
2x ' ^pAzzword
```



9. Based on the decoded text: “**steghide**”, I used the following command:

```
(kali@kali)-[~/Downloads]
$ steghide extract -sf img.jpg
Enter passphrase:
wrote extracted data to "flag.txt".

(kali@kali)-[~/Downloads]
$ █
```

steghide extract -sf img.jpg

10. When prompted for a password, entered **pAzzword**. (You can't see it)

```
Enter passphrase:
wrote extracted data to "fl
```

11. The hidden file named flag.txt was successfully extracted, revealing the challenge flag.

```
(kali@kali)-[~/Downloads]
$ steghide extract -sf img.jpg
Enter passphrase:
wrote extracted data to "flag.txt".
Encryption / Encoding
(kali@kali)-[~/Downloads]
$ cat flag.txt
picoCTF{h1dd3n_1n_1m4g3_2ac27d95}

(kali@kali)-[~/Downloads]
$ █
```



V. Result

picoCTF{h1dd3n_1n_1m4g3_2ac27d95}

```
(kali@kali)-[~/Downloads]
$ cat flag.txt
picoCTF{h1dd3n_1n_1m4g3_2ac27d95}
```

VI. Explanation

This challenge demonstrates how information can be hidden inside images through **steganography**.

The investigation began by analyzing the image metadata using **ExifTool**, which revealed a suspicious encoded string in the comments field. Encoded data is often used in challenges to hide clues, so the string was decoded using **Base64**.

The decoded result provided two important hints:

- **steghide** – indicating the tool required to extract hidden content
- **pAzzword** – the password needed for extraction

Explanation of the command:

steghide → A tool used for hiding or extracting data within files using steganography.

extract → Retrieves hidden data from the stego file.

-sf → Specifies the **stego file** (the file containing hidden data).

img.jpg → The image file containing the embedded data.



Using the **steghide extract** command allowed the hidden data embedded within the image to be recovered. After entering the correct password, the concealed file was extracted and the flag was revealed.

This process highlights the importance of examining both **metadata** and **hidden file structures** when performing forensic analysis.

VII. Conclusion

The Hidden in plainsight challenge illustrates how steganography and encoded metadata can be used to conceal information within image files. By analyzing metadata, decoding encoded strings, and using steganography tools, the hidden data was successfully extracted. This challenge emphasizes the importance of combining forensic analysis techniques with decoding and specialized tools to uncover hidden information.

— **The Analyst: Hyposelenia**